Predicting Ozone Nonattainment in US Counties with Supervised Learning

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1. Problem statement

Ground-level ozone is a regulated pollutant. A US county is designated **nonattainment** when its ozone design value (3-year average of the annual 4th-highest daily maximum at the worst monitor) exceeds 0.070 ppm, which triggers costly mandated controls. We predict daily ozone from weather and precursor pollutants, then roll predictions up to **county nonattainment status and severity tier**, so regions can act before designation. Our work extends the CS229 (2019) ozone-regression report, fixing the weaknesses it listed.

2. Dataset

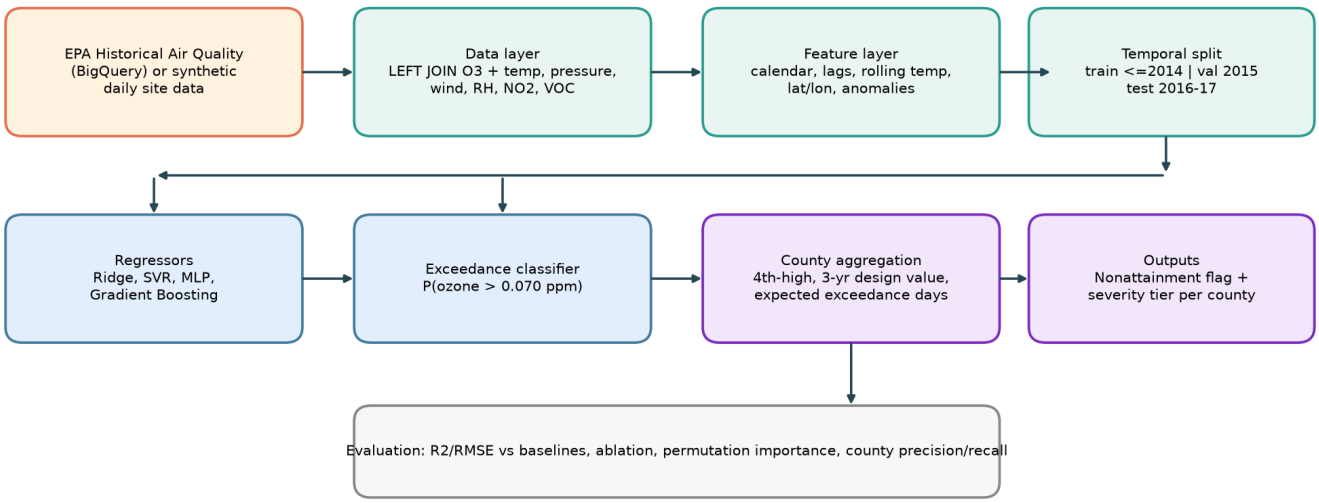
Source: EPA Historical Air Quality daily summaries (Google BigQuery public data, 2000-2017): ozone, temperature, pressure, wind, relative humidity, NO_2, VOC, joined by site and day with LEFT JOINs. The numbers below were produced on **EPA Historical Air Quality data**: 5,423,797 train / 381,166 validation / 777,669 test site-days, 1965 sites, 934 counties. Sites missing a sensor (NO_2, VOC, wind, ...) are kept and imputed rather than dropped. Reproduce with `python train.py --source bigquery`.

3. Approach

Fixes over the paper: (i) time-based split by whole calendar years (train ≤2014, validation 2015, test 2016-17) instead of a Jan-Jun / Jul-Dec seasonal split; (ii) standard $R^2 = 1 - SSE/SST$ (the paper divided by Σy^2); (iii) latitude/longitude instead of a numeric site ID; (iv) missing sensors kept via median imputation plus missing-indicators or native NaN handling, instead of dropping rows; (v) SVR target scaled so ε is meaningful and tuned (paper's ε was too large, giving negative R^2); (vi) naive baselines and a no-persistence ablation; (vii) permutation feature importance; (viii) a county-level classifier with severity tiers.

Models: Ridge regression, RBF support vector regression, MLP (128-128-64, ReLU, adaptive learning rate, early stopping), histogram gradient boosting, plus persistence and site-month climatology baselines. The best model on validation RMSE (hgb) is used downstream. A gradient-boosting classifier gives P(daily ozone > 0.070 ppm); summing probabilities per site-year gives expected exceedance days, and ≥4 means the 4th-highest day exceeds the standard. Regression outputs are rolled into annual 4th-high values, bias-calibrated on the validation year, averaged over 3 years into design values, and binned into Marginal-Extreme tiers.

System architecture



4. Implementation overview

Python (pandas, scikit-learn, matplotlib). `src/data_bigquery.py` and `data_synthetic.py` load data; `features.py` builds leakage-safe features (lags use only past days); `models.py` holds the model zoo; `county.py` implements EPA design-value logic; `train.py` runs the full pipeline and writes metrics, plots and a saved model; `demo.py` answers what-if questions (e.g. +4°F) per county; 6 pytest tests cover splits, lag leakage, design values and model sanity.

5. Results (test years 2016-2017)

Model	Val R2	Test R2	Test RMSE (ppm)	Test MAE (ppm)
persistence	0.500	0.482	0.0089	0.0066
climatology	0.327	0.322	0.0102	0.0079

ridge	0.592	0.579	0.0080	0.0062
svr	0.614	0.603	0.0078	0.0059
mlp	0.630	0.605	0.0078	0.0059
hgb	0.634	0.632	0.0075	0.0057

Without yesterday's ozone the gradient-boosting model still reaches R^2 = 0.476 using meteorology and precursors alone. Daily exceedance classifier: ROC-AUC 0.971, PR-AUC 0.532, F1 0.529 (exceedance base rate 1.4%). County-year nonattainment (n=2370): regression-based accuracy 0.867, F1 0.667; exceedance-count method accuracy 0.870, F1 0.593. 2017 design values: accuracy 0.869, MAE 0.0026 ppm, severity tier exact 0.86 / within one tier 1.00.

6. Conclusions and limitations

HGB was the best model on validation RMSE (test R^2 0.632, RMSE 0.0075 ppm), compared with Ridge (0.579) and the persistence baseline (0.482). Rolling daily predictions into county design values flags nonattainment with 87% accuracy and places counties within one severity tier 100% of the time, but nonattainment recall is 59%, so some violating counties are missed. Without yesterday's ozone the model drops to R^2 0.48, showing that weather and NO_2/VOC alone explain only part of daily ozone. **Limitations:** the daily target is the 1-hour maximum rather than the regulatory 8-hour average, so 'nonattainment' here is a proxy; the SVR was trained on a 15,000-row subsample because it scales quadratically; the MLP was stopped early to save time on the large training set, so its score is a lower bound; completeness rules and exceptional-event exclusions are not modelled; tier cut-offs are simplified; emission-source features (factories, highways, census data) and spatial/temporal deep models are future work.